CS-300 Analysis and Design

Project One Milestone Two: Hash Table Data

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Overview

This pseudocode outlines the process for loading course data from a file, validating the data, creating course objects, storing them in a hash table, and printing course information. The design follows the requirements provided by ABC University and prepares the foundation for Project One.

A1: Open File and Read Lines

```
BEGIN
    PROMPT user for course data file path
    OPEN file at given path

    IF file cannot be opened
        PRINT “Error: Cannot open file.”
        STOP program
    ENDIF
```

A2: First Pass: Collect All Course Numbers

```
CREATE empty HashTable called courseTable
CREATE empty list called allCourseNumbers

FOR each line in file
    SPLIT line by commas into tokens
    IF number of tokens < 2
        PRINT "Format Error: Each line must contain at least a course number and a
        course name."
```

```
            CONTINUE to next line
        ENDIF

        SET courseNumber = tokens[0]
        ADD courseNumber to allCourseNumbers
ENDFOR
```

A3: Second Pass: Validate Prerequisites and Create Course Objects

```
RESET file pointer to beginning of file
FOR each line in file
        SPLIT line by commas into tokens
        SET courseNumber = tokens[0]
        SET courseName   = tokens[1]
        CREATE empty list prereqList
        FOR each token after the second token
                SET prereq = token
                IF prereq NOT IN allCourseNumbers
                        PRINT "Error: Prerequisite " + prereq + " does not exist in file."
                        CONTINUE to next line
                ENDIF
                ADD prereq to prereqList
        ENDFOR

        CREATE new Course object
        SET Course.number = courseNumber
```

```
        SET Course.name   = courseName
        SET Course.prerequisites = prereqList

        INSERT Course object into courseTable using courseNumber as key
    ENDFOR
END
```

B1: Course Structure Definition

```
STRUCT Course
        STRING number
        STRING name
        LIST prerequisites
END STRUCT
```

C1: Insert Course Into Hash Table

```
FUNCTION insertCourse(courseTable, course)
        COMPUTE key = hash(course.number)
        IF bucket at key is empty
                STORE course in bucket
        ELSE
                CHAIN course to end of linked list at bucket
        ENDIF
END FUNCTION
```

D1: Print a Single Course And Its Prerequisites

```
FUNCTION printCourse(courseTable, courseNumber)
    COMPUTE key = hash(courseNumber)
    SET currentNode = bucket at key

    WHILE currentNode is not null
        IF currentNode.course.number == courseNumber

            PRINT currentNode.course.number + ": " + currentNode.course.name

            IF currentNode.course.prerequisites list is empty
                PRINT " No prerequisites."
            ELSE
                PRINT " Prerequisites:"
                FOR each prereq in currentNode.course.prerequisites
                    PRINT "    " + prereq
                ENDFOR
            ENDIF
            RETURN
        ENDIF
        SET currentNode = currentNode.next
    ENDWHILE
    PRINT "Course " + courseNumber + " not found."
END FUNCTION
```

D2: Print All Courses In The Hash Tables

```
FUNCTION printAllCourses(courseTable)
    FOR each bucket in courseTable
        SET currentNode = bucket
        WHILE currentNode is not null
            PRINT currentNode.course.number + ": " + currentNode.course.name
            IF currentNode.course.prerequisites list is empty
                PRINT " No prerequisites."
            ELSE
                PRINT " Prerequisities:"
                FOR each prereq in currentNode.course.prerequisites
                    PRINT "    " + prereq
                ENDFOR
            ENDIF
            SET currentNode = currentNode.next
        ENDWHILE
    ENDFOR
END FUNCTION
```